CC3D Matrix-Degradation Calibration Analysis Report

Old-code reference vs. new MMP-only implementation | First-pass results

1. Executive summary

Eight new-code parameter values (var = 0.25, 0.5, 1.0, 1.5, 2.0, 2.5, 3.0, 3.5) were compared with the old-code reference at var = 1.5. The available outputs contain one seed (12345) and one arrangement (A) per new-code value. In this first-pass comparison, var = 1.0 has the lowest overall score (0.1646), followed by 0.5 (0.1859) and 0.25 (0.1975). Lower scores indicate greater similarity to the old-code reference under the analysis script's weighted metric.

This is a provisional ranking, not a final calibration decision. COI spatial overlap is the metric that most clearly distinguishes the tested values; laminin overlap is almost perfect for all candidates. The analysis currently uses the common final MCS (4439) for the radial score and the final overlay, so those outputs should be regenerated with the planned MCS 3000 spatial checkpoint to reduce possible lattice-boundary influence. Temporal curves should still use the full run.

2. Simulation comparison and score interpretation

Reference: code_version = old, var = 1.5. Candidates: code_version = new, all tested var values. A candidate is compared only to a reference with the same arrangement_id and matrix_signature; the same seed is paired when available. The current rows show arrangement A, seed 12345, one matching reference.

The script's per-component score (lower is better) is:

- 0.6 × (1 − mean spatial IoU across checkpoints)
- 0.2 × min(1, temporal RMSE of cumulative degraded-% curves / 100)
- 0.1 × mean absolute radial-bin degradation-fraction difference at the evaluation MCS
- 0.1 × min(1, absolute first-degradation-time difference / evaluation MCS)

The overall score is the average of COI and laminin component scores (w_coi = 0.5). Spatial IoU measures overlap between the sets of baseline matrix cells that have experienced a degradation event by a checkpoint. Precision and recall are also reported in summary_by_run.csv. Because events are cumulative, a matrix cell remains counted as degraded after its first logged degradation event, even if its cell state later changes.

3. Results by parameter value

The table below is taken from summary_by_var.csv. IoU values are the means over the checkpoints used in this run (the default checkpoint schedule extended to the common last MCS, 4439).

Rank	New var	Overall score ↓	COI IoU ↑	Laminin IoU ↑	COI curve RMSE (percentage points)
1	1	0.1646	0.485	1.000	1.80
2	0.5	0.1859	0.418	1.000	1.60
3	0.25	0.1975	0.390	1.000	2.15
4	1.5	0.2243	0.303	1.000	5.14
5	2	0.2428	0.243	1.000	6.12
6	3	0.2838	0.159	0.983	7.29
7	2.5	0.2990	0.089	0.996	8.09
8	3.5	0.3086	0.077	0.996	8.21

4. What the current metrics say

4.1 COI spatial overlap

var = 1.0 has the highest mean COI IoU (0.485), followed by 0.5 (0.418) and 0.25 (0.390). At larger values the overlap generally deteriorates: 1.5 = 0.303, 2.0 = 0.243, 2.5 = 0.089, 3.0 = 0.159, and 3.5 = 0.077. This supports focusing the next comparison on var = 0.5 and 1.0, while retaining 0.25 as a useful lower-value comparator.

4.2 Laminin spatial overlap

Laminin IoU is 1.000 for var = 0.25, 0.5, 1.0, 1.5, and 2.0; it remains 0.983–0.996 for the higher values. Thus, in this run, laminin spatial overlap does not strongly discriminate among most candidates. It is still important to retain laminin temporal and onset measures because matching the final spatial set alone does not imply matching the timing of degradation.

4.3 COI timing and final degraded fraction

Run	First COI degradation (MCS)	COI degraded at end	Difference vs old endpoint	Interpretation
Old reference	35	13.10%	Reference	Old-code endpoint
New var=0.25	8	14.29%	+1.19 pp	
New var=0.5	18	11.51%	-1.59 pp	
New var=1	24	10.12%	-2.98 pp	
New var=1.5	44	5.16%	-7.94 pp	
New var=2	92	3.77%	-9.33 pp	
New var=2.5	268	1.98%	-11.11 pp	
New var=3	228	2.18%	-10.91 pp	
New var=3.5	288	1.19%	-11.90 pp	

The old reference first degrades COI at MCS 35 and has 13.10% degraded by the common endpoint. New var = 0.25 starts much earlier (MCS 8) and ends at 14.29%, so its endpoint fraction is closest to the reference but its timing differs substantially. var = 0.5 starts at MCS 18 and ends at 11.51%. var = 1.0 starts at MCS 24 and ends at 10.12%; it has the best spatial IoU but underestimates the old endpoint fraction by about 2.98 percentage points. New var = 1.5 starts at MCS 44 and ends at 5.16%, which is substantially below the old reference.

4.4 Temporal curves

COI cumulative degraded-percentage RMSE is lowest for var = 0.5 (1.60 percentage points), followed by var = 1.0 (1.80) and var = 0.25 (2.15). This means var = 0.5 matches the COI time course slightly better than var = 1.0, even though var = 1.0 has the better spatial IoU and overall score. Laminin RMSE is lowest at var = 2.0 (1.73), then var = 1.5 (3.67) and 2.5 (3.76); the best parameter can therefore differ by component and metric.

4.5 MMP event distributions

The old-code MMP values recorded at degradation events have the following percentiles (5th / 25th / median / 75th / 95th):

- COI: 0.106 / 0.4461 / 0.9257 / 1.624 / 2.469
- Laminin: 1.122 / 1.65 / 2.453 / 3.29 / 4.162

These percentiles describe the MMP values associated with old-code degradation events; they are not by themselves a direct mapping from old var to the correct new var. The maximum MMP at intact COI COM observed in the new runs ranges from 1.122 (var = 0.25) to 4.555 (var = 3.0), with var = 1.5 at 3.309. Treat these maxima as descriptive diagnostics, not a calibration score.

5. Recommended next analysis

1. Set the spatial overlay snapshot to MCS 3000 instead of the common final MCS 4439.
2. Explicitly include MCS 3000 in the spatial-IoU checkpoints. Keep additional earlier checkpoints so the result is not based on one image alone.

3. Evaluate the radial score at MCS 3000 as well, because the current radial score uses the common final MCS.
4. Keep temporal curves and temporal RMSE over the full common simulation duration; these describe kinetics rather than only one snapshot.
5. Compare var = 0.5 and var = 1.0 closely; keep var = 0.25 as a lower-value comparator. Do not call var = 1.0 final until the MCS 3000 spatial comparison and repeat seeds are assessed.
6. Run additional seeds for the leading values and, ideally, repeat the old reference too. The current standard deviation is NaN because there is only one seed per value; seed-to-seed variability cannot yet be estimated.

Important code note: in the current analysis script, --checkpoints controls the checkpoints used for the IoU score, but the overlay plot itself uses tend (the common final MCS). Changing only --checkpoints will not move the overlay to MCS 3000. The overlay code must use a separate overlay_mcs = 3000 value, and the radial metric should likewise use its intended evaluation time.

6. What each analysis output file tells us

summary_by_run.csv

- One row per candidate run. Includes arrangement, var, seed, whether the reference seed was paired, number of references used, overall score, component-specific scores, COI and laminin IoU/precision/recall, temporal RMSE, final degraded percentages, radial difference, and first degradation MCS. Use this to inspect the detailed trade-offs for an individual run.

summary_by_var.csv

- Aggregates candidate runs by arrangement and var. Reports number of seeds, mean overall score, standard deviation, mean COI/laminin IoU, temporal RMSE, and rank within the arrangement. Use this for parameter ranking. With one seed, the SD is unavailable (NaN), and the rank is preliminary.

suggested_var_range.txt

- Reports percentiles of old-code MMP at degradation events and maximum MMP at intact COI COM for each run. Use as a diagnostic guide to the MMP range represented in the simulations; it does not automatically choose the optimal new var.

curve_COI_arrA.png

- Plots cumulative percentage of initial COI matrix cells that have experienced degradation against MCS for each new var, alongside the old-code reference. Useful for timing and total-extent comparisons.

curve_laminin_arrA.png

- Same type of cumulative degradation curve for laminin. Useful for seeing whether degradation starts earlier/later and how quickly the cumulative fraction grows.

score_vs_var_arrA.png

- Plots overall score against new-code var; lower is better. Error bars reflect across-seed standard deviation where multiple seeds exist. Here only one seed is present per value, so there is no meaningful estimated seed variability.

overlay_best_arrA.png

- Two-panel spatial map for COI and laminin showing cells degraded in both old and new runs, old-only, new-only, and neither, with the cancer centroid marked. In the current script this overlay is generated at tend (4439), not at the requested MCS 3000; update the code before using it to judge pre-boundary spatial agreement.

7. What each logger file provides

_meta.json

- Run metadata: code version, var, arrangement, seed, matrix signature, final MCS and baseline centroid. Used to classify old vs new, pair runs, and check whether initial layouts are comparable.

_rows.csv

- Time-series log at the logger's sampling interval, including MCS, component degradation percentages and MMP diagnostics. Used for temporal curves and maximum-MMP summaries.

_events.csv

- Event-level records of degradation: component, cell ID, MCS, whether the cell was part of the baseline, and MMP at the event. Used to reconstruct cumulative degraded-cell sets and event-time distributions.

_baseline_cells.csv

- One row per baseline matrix cell, with cell ID, component, initial position and initial area. Defines the cells used as the comparison denominator and spatial unit.

_baseline_maps.npz

- Saved baseline lattice maps (labels/types and matrix-related maps). Useful for reconstructing or visually checking the initial spatial layout; the current analysis script primarily uses baseline cell IDs and positions from the CSV.

_types_mcsNNNNNN.npz (if enabled)

- Optional lattice type/label snapshots at selected MCS values. These can support visual checks of the evolving lattice, but the analysis outputs described above are generated from the event and baseline logs.

8. Caveats and interpretation

- This report describes the supplied output files and the current script logic; it does not claim that the MCS 3000 rerun has already been performed.
- The current analysis uses one seed (12345) and arrangement A, so differences could partly reflect stochastic variation.
- The near-perfect laminin IoU should be read alongside its timing metrics; spatial overlap alone can conceal differences in when degradation occurs.
- A low composite score is only as appropriate as its chosen weights and evaluation times. Since spatial pattern is a primary scientific criterion, inspect component-specific IoU and overlays alongside the overall score.
- The CSV's final percentages refer to the common last MCS used in this analysis, 4439. They should not be interpreted as MCS 3000 values.

9. Current takeaway

For the current data, var = 1.0 is the best composite-score candidate and has the strongest COI spatial overlap, while var = 0.5 gives the lowest COI temporal RMSE and var = 0.25 is closest to the old endpoint percentage. The next defensible step is to revise the spatial overlay and radial evaluation to MCS 3000, retain full-run temporal analysis, and repeat the leading values across multiple seeds before choosing a final calibration.